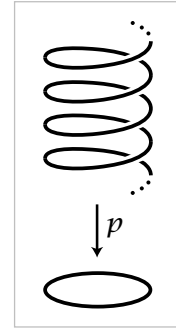


The Fundamental Group of the Circle

Our first real theorem will be the calculation $\pi_1(S^1) \approx \mathbb{Z}$. Besides its intrinsic interest, this basic result will have several immediate applications of some substance, and it will be the starting point for many more calculations in the next section. It should be no surprise then that the proof will involve some genuine work. To maximize the payoff for this work, the proof is written so that its main technical steps apply in the more general setting of covering spaces, the main topic of §1.3.

Theorem 1.7. *The map $\Phi: \mathbb{Z} \rightarrow \pi_1(S^1)$ sending an integer n to the homotopy class of the loop $\omega_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$ based at $(1, 0)$ is an isomorphism.*

Proof: The idea is to compare paths in S^1 with paths in $\mathbb{R}$ via the map $p: \mathbb{R} \rightarrow S^1$ given by $p(s) = (\cos 2\pi s, \sin 2\pi s)$. This map can be visualized geometrically by embedding $\mathbb{R}$ in $\mathbb{R}^3$ as the helix parametrized by $s \mapsto (\cos 2\pi s, \sin 2\pi s, s)$, and then p is the restriction to the helix of the projection of $\mathbb{R}^3$ onto $\mathbb{R}^2$, $(x, y, z) \mapsto (x, y)$, as in the figure. Observe that the loop ω_n is the composition $p\tilde{\omega}_n$ where $\tilde{\omega}_n: I \rightarrow \mathbb{R}$ is the path $\tilde{\omega}_n(s) = ns$, starting at 0 and ending at n , winding around the helix $|n|$ times, upward if $n > 0$ and downward if $n < 0$. The relation $\omega_n = p\tilde{\omega}_n$ is expressed by saying that $\tilde{\omega}_n$ is a **lift** of ω_n .



The definition of Φ can be reformulated by setting $\Phi(n)$ equal to the homotopy class of the loop $p\tilde{f}$ for $\tilde{f}$ any path in $\mathbb{R}$ from 0 to n . Such an $\tilde{f}$ is homotopic to $\tilde{\omega}_n$ via the linear homotopy $(1-t)\tilde{f} + t\tilde{\omega}_n$, hence $p\tilde{f}$ is homotopic to $p\tilde{\omega}_n = \omega_n$ and the new definition of $\Phi(n)$ agrees with the old one.

To verify that Φ is a homomorphism, let $\tau_m: \mathbb{R} \rightarrow \mathbb{R}$ be the translation $\tau_m(x) = x + m$. Then $\tilde{\omega}_m \cdot (\tau_m\tilde{\omega}_n)$ is a path in $\mathbb{R}$ from 0 to $m + n$, so $\Phi(m + n)$ is the homotopy class of the loop in S^1 that is the image of this path under p . This image is just $\omega_m \cdot \omega_n$, so $\Phi(m + n) = \Phi(m) \cdot \Phi(n)$.

To show that Φ is an isomorphism we shall use two facts:

- (a) For each path $f: I \rightarrow S^1$ starting at a point $x_0 \in S^1$ and each $\tilde{x}_0 \in p^{-1}(x_0)$ there is a unique lift $\tilde{f}: I \rightarrow \mathbb{R}$ starting at $\tilde{x}_0$.
- (b) For each homotopy $f_t: I \rightarrow S^1$ of paths starting at x_0 and each $\tilde{x}_0 \in p^{-1}(x_0)$ there is a unique lifted homotopy $\tilde{f}_t: I \rightarrow \mathbb{R}$ of paths starting at $\tilde{x}_0$.

Before proving these facts, let us see how they imply the theorem. To show that Φ is surjective, let $f: I \rightarrow S^1$ be a loop at the basepoint $(1, 0)$, representing a given element of $\pi_1(S^1)$. By (a) there is a lift $\tilde{f}$ starting at 0. This path $\tilde{f}$ ends at some integer n since $p\tilde{f}(1) = f(1) = (1, 0)$ and $p^{-1}(1, 0) = \mathbb{Z} \subset \mathbb{R}$. By the extended definition of Φ we then have $\Phi(n) = [p\tilde{f}] = [f]$. Hence Φ is surjective.

To show that Φ is injective, suppose $\Phi(m) = \Phi(n)$, which means $\omega_m \simeq \omega_n$. Let f_t be a homotopy from $\omega_m = f_0$ to $\omega_n = f_1$. By (b) this homotopy lifts to a homotopy $\tilde{f}_t$ of paths starting at 0. The uniqueness part of (a) implies that $\tilde{f}_0 = \tilde{\omega}_m$ and $\tilde{f}_1 = \tilde{\omega}_n$. Since $\tilde{f}_t$ is a homotopy of paths, the endpoint $\tilde{f}_t(1)$ is independent of t . For $t = 0$ this endpoint is m and for $t = 1$ it is n , so $m = n$.

It remains to prove (a) and (b). Both statements can be deduced from a more general assertion:

- (c) Given a map $F: Y \times I \rightarrow S^1$ and a map $\tilde{F}: Y \times \{0\} \rightarrow \mathbb{R}$ lifting $F|_{Y \times \{0\}}$, then there is a unique map $\tilde{F}: Y \times I \rightarrow \mathbb{R}$ lifting F and restricting to the given $\tilde{F}$ on $Y \times \{0\}$.

Statement (a) is the special case that Y is a point, and (b) is obtained by applying (c) with $Y = I$ in the following way. The homotopy f_t in (b) gives a map $F: I \times I \rightarrow S^1$ by setting $F(s, t) = f_t(s)$ as usual. A unique lift $\tilde{F}: I \times \{0\} \rightarrow \mathbb{R}$ is obtained by an application of (a). Then (c) gives a unique lift $\tilde{F}: I \times I \rightarrow \mathbb{R}$. The restrictions $\tilde{F}|_{\{0\} \times I}$ and $\tilde{F}|_{\{1\} \times I}$ are paths lifting constant paths, hence they must also be constant by the uniqueness part of (a). So $\tilde{f}_t(s) = \tilde{F}(s, t)$ is a homotopy of paths, and $\tilde{f}_t$ lifts f_t since $p\tilde{F} = F$.

We shall prove (c) using just one special property of the projection $p: \mathbb{R} \rightarrow S^1$, namely:

- There is an open cover $\{U_\alpha\}$ of S^1 such that for each α , $p^{-1}(U_\alpha)$ can be
 (*) decomposed as a disjoint union of open sets each of which is mapped homeomorphically onto U_α by p .

For example, we could take the cover $\{U_\alpha\}$ to consist of any two open arcs in S^1 whose union is S^1 .

To prove (c) we will first construct a lift $\tilde{F}: N \times I \rightarrow \mathbb{R}$ for N some neighborhood in Y of a given point $y_0 \in Y$. Since F is continuous, every point $(y_0, t) \in Y \times I$ has a product neighborhood $N_t \times (a_t, b_t)$ such that $F(N_t \times (a_t, b_t)) \subset U_\alpha$ for some α . By compactness of $\{y_0\} \times I$, finitely many such products $N_t \times (a_t, b_t)$ cover $\{y_0\} \times I$. This implies that we can choose a single neighborhood N of y_0 and a partition $0 = t_0 < t_1 < \cdots < t_m = 1$ of I so that for each i , $F(N \times [t_i, t_{i+1}])$ is contained in some U_α , which we denote U_i . Assume inductively that $\tilde{F}$ has been constructed on $N \times [0, t_i]$. We have $F(N \times [t_i, t_{i+1}]) \subset U_i$, so by (*) there is an open set $\tilde{U}_i \subset \mathbb{R}$ projecting homeomorphically onto U_i by p and containing the point $\tilde{F}(y_0, t_i)$. After replacing N by a smaller neighborhood of y_0 we may assume that $\tilde{F}(N \times \{t_i\})$ is contained in $\tilde{U}_i$, namely, replace $N \times \{t_i\}$ by its intersection with $(\tilde{F}|_{N \times \{t_i\}})^{-1}(\tilde{U}_i)$. Now we can define $\tilde{F}$ on $N \times [t_i, t_{i+1}]$ to be the composition of F with the homeomorphism $p^{-1}: U_i \rightarrow \tilde{U}_i$. After finitely many repetitions of this induction step we eventually get a lift $\tilde{F}: N \times I \rightarrow \mathbb{R}$ for some neighborhood N of y_0 .

Next we show the uniqueness part of (c) in the special case that Y is a point. In this case we can omit Y from the notation. So suppose $\tilde{F}$ and $\tilde{F}'$ are two lifts of $F: I \rightarrow S^1$

such that $\tilde{F}(0) = \tilde{F}'(0)$. As before, choose a partition $0 = t_0 < t_1 < \cdots < t_m = 1$ of I so that for each i , $F([t_i, t_{i+1}])$ is contained in some U_i . Assume inductively that $\tilde{F} = \tilde{F}'$ on $[0, t_i]$. Since $[t_i, t_{i+1}]$ is connected, so is $\tilde{F}([t_i, t_{i+1}])$, which must therefore lie in a single one of the disjoint open sets $\tilde{U}_i$ projecting homeomorphically to U_i as in (*). By the same token, $\tilde{F}'([t_i, t_{i+1}])$ lies in a single $\tilde{U}_i$, in fact in the same one that contains $\tilde{F}([t_i, t_{i+1}])$ since $\tilde{F}'(t_i) = \tilde{F}(t_i)$. Because p is injective on $\tilde{U}_i$ and $p\tilde{F} = p\tilde{F}'$, it follows that $\tilde{F} = \tilde{F}'$ on $[t_i, t_{i+1}]$, and the induction step is finished.

The last step in the proof of (c) is to observe that since the $\tilde{F}$'s constructed above on sets of the form $N \times I$ are unique when restricted to each segment $\{y\} \times I$, they must agree whenever two such sets $N \times I$ overlap. So we obtain a well-defined lift $\tilde{F}$ on all of $Y \times I$. This $\tilde{F}$ is continuous since it is continuous on each $N \times I$, and it is unique since it is unique on each segment $\{y\} \times I$. $\square$

Now we turn to some applications of this theorem. Although algebraic topology is usually 'algebra serving topology,' the roles are reversed in the following proof of the Fundamental Theorem of Algebra.

Theorem 1.8. *Every nonconstant polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.*

Proof: We may assume the polynomial is of the form $p(z) = z^n + a_1 z^{n-1} + \cdots + a_n$. If $p(z)$ has no roots in $\mathbb{C}$, then for each real number $r \geq 0$ the formula

$$f_r(s) = \frac{p(re^{2\pi i s})/p(r)}{|p(re^{2\pi i s})/p(r)|}$$

defines a loop in the unit circle $S^1 \subset \mathbb{C}$ based at 1. As r varies, f_r is a homotopy of loops based at 1. Since f_0 is the trivial loop, we deduce that the class $[f_r] \in \pi_1(S^1)$ is zero for all r . Now fix a large value of r , bigger than $|a_1| + \cdots + |a_n|$ and bigger than 1. Then for $|z| = r$ we have

$$|z^n| = r^n = r \cdot r^{n-1} > (|a_1| + \cdots + |a_n|)|z^{n-1}| \geq |a_1 z^{n-1} + \cdots + a_n|$$

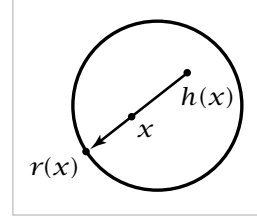
From the inequality $|z^n| > |a_1 z^{n-1} + \cdots + a_n|$ it follows that the polynomial $p_t(z) = z^n + t(a_1 z^{n-1} + \cdots + a_n)$ has no roots on the circle $|z| = r$ when $0 \leq t \leq 1$. Replacing p by p_t in the formula for f_r above and letting t go from 1 to 0, we obtain a homotopy from the loop f_r to the loop $\omega_n(s) = e^{2\pi i n s}$. By Theorem 1.7, ω_n represents n times a generator of the infinite cyclic group $\pi_1(S^1)$. Since we have shown that $[\omega_n] = [f_r] = 0$, we conclude that $n = 0$. Thus the only polynomials without roots in $\mathbb{C}$ are constants. $\square$

Our next application is the Brouwer fixed point theorem in dimension 2.

Theorem 1.9. *Every continuous map $h: D^2 \rightarrow D^2$ has a fixed point, that is, a point x with $h(x) = x$.*

Here we are using the standard notation D^n for the closed unit disk in $\mathbb{R}^n$, all vectors x of length $|x| \leq 1$. Thus the boundary of D^n is the unit sphere S^{n-1} .

Proof: Suppose on the contrary that $h(x) \neq x$ for all $x \in D^2$. Then we can define a map $r: D^2 \rightarrow S^1$ by letting $r(x)$ be the point of S^1 where the ray in $\mathbb{R}^2$ starting at $h(x)$ and passing through x leaves D^2 . Continuity of r is clear since small perturbations of x produce small perturbations of $h(x)$, hence also small perturbations of the ray through these two points.



The crucial property of r , besides continuity, is that $r(x) = x$ if $x \in S^1$. Thus r is a retraction of D^2 onto S^1 . We will show that no such retraction can exist.

Let f_0 be any loop in S^1 . In D^2 there is a homotopy of f_0 to a constant loop, for example the linear homotopy $f_t(s) = (1 - t)f_0(s) + tx_0$ where x_0 is the basepoint of f_0 . Since the retraction r is the identity on S^1 , the composition rf_t is then a homotopy in S^1 from $rf_0 = f_0$ to the constant loop at x_0 . But this contradicts the fact that $\pi_1(S^1)$ is nonzero. $\square$

This theorem was first proved by Brouwer around 1910, one of the early triumphs of algebraic topology. Brouwer in fact proved the corresponding result for D^n , and we shall obtain this generalization in Corollary 2.11 using homology groups in place of π_1 . One could also use the higher homotopy group π_n . Brouwer's original proof used neither homology nor homotopy groups, which had not been invented at the time. Instead it used the notion of degree for maps $S^n \rightarrow S^n$, which we shall define in §2.2 using homology but which Brouwer defined directly in more geometric terms.

These proofs are all arguments by contradiction, and so they show just the existence of fixed points without giving any clue as to how to find one in explicit cases. Our proof of the Fundamental Theorem of Algebra was similar in this regard. There exist other proofs of the Brouwer fixed point theorem that are somewhat more constructive, for example the elegant and quite elementary proof by Sperner in 1928, which is explained very nicely in [Aigner-Ziegler 1999].

The techniques used to calculate $\pi_1(S^1)$ can be applied to prove the Borsuk-Ulam theorem in dimension two:

Theorem 1.10. *For every continuous map $f: S^2 \rightarrow \mathbb{R}^2$ there exists a pair of antipodal points x and $-x$ in S^2 with $f(x) = f(-x)$.*

It may be that there is only one such pair of antipodal points $x, -x$, for example if f is simply orthogonal projection of the standard sphere $S^2 \subset \mathbb{R}^3$ onto a plane.

The Borsuk-Ulam theorem holds also for maps $S^n \rightarrow \mathbb{R}^n$, as we show in Proposition 2B.6. The proof for $n = 1$ is easy since the difference $f(x) - f(-x)$ changes sign as x goes halfway around the circle, hence this difference must be zero for some x . For $n \geq 2$ the theorem is certainly less obvious. Is it apparent, for example, that at every instant there must be a pair of antipodal points on the surface of the earth having the same temperature and the same barometric pressure?

The theorem says in particular that there is no one-to-one continuous map from S^2 to $\mathbb{R}^2$, so S^2 is not homeomorphic to a subspace of $\mathbb{R}^2$, an intuitively obvious fact that is not easy to prove directly.

Proof: If the conclusion is false for $f: S^2 \rightarrow \mathbb{R}^2$, we can define a map $g: S^2 \rightarrow S^1$ by $g(x) = (f(x) - f(-x)) / |f(x) - f(-x)|$. Define a loop η circling the equator of $S^2 \subset \mathbb{R}^3$ by $\eta(s) = (\cos 2\pi s, \sin 2\pi s, 0)$, and let $h: I \rightarrow S^1$ be the composed loop $g\eta$. Since $g(-x) = -g(x)$, we have the relation $h(s + 1/2) = -h(s)$ for all s in the interval $[0, 1/2]$. As we showed in the calculation of $\pi_1(S^1)$, the loop h can be lifted to a path $\tilde{h}: I \rightarrow \mathbb{R}$. The equation $h(s + 1/2) = -h(s)$ implies that $\tilde{h}(s + 1/2) = \tilde{h}(s) + q/2$ for some odd integer q that might conceivably depend on $s \in [0, 1/2]$. But in fact q is independent of s since by solving the equation $\tilde{h}(s + 1/2) = \tilde{h}(s) + q/2$ for q we see that q depends continuously on $s \in [0, 1/2]$, so q must be a constant since it is constrained to integer values. In particular, we have $\tilde{h}(1) = \tilde{h}(1/2) + q/2 = \tilde{h}(0) + q$. This means that h represents q times a generator of $\pi_1(S^1)$. Since q is odd, we conclude that h is not nullhomotopic. But h was the composition $g\eta: I \rightarrow S^2 \rightarrow S^1$, and η is obviously nullhomotopic in S^2 , so $g\eta$ is nullhomotopic in S^1 by composing a nullhomotopy of η with g . Thus we have arrived at a contradiction. $\square$

Corollary 1.11. *Whenever S^2 is expressed as the union of three closed sets A_1, A_2 , and A_3 , then at least one of these sets must contain a pair of antipodal points $\{x, -x\}$.*

Proof: Let $d_i: S^2 \rightarrow \mathbb{R}$ measure distance to A_i , that is, $d_i(x) = \inf_{y \in A_i} |x - y|$. This is a continuous function, so we may apply the Borsuk-Ulam theorem to the map $S^2 \rightarrow \mathbb{R}^2$, $x \mapsto (d_1(x), d_2(x))$, obtaining a pair of antipodal points x and $-x$ with $d_1(x) = d_1(-x)$ and $d_2(x) = d_2(-x)$. If either of these two distances is zero, then x and $-x$ both lie in the same set A_1 or A_2 since these are closed sets. On the other hand, if the distances from x and $-x$ to A_1 and A_2 are both strictly positive, then x and $-x$ lie in neither A_1 nor A_2 so they must lie in A_3 . $\square$

To see that the number ‘three’ in this result is best possible, consider a sphere inscribed in a tetrahedron. Projecting the four faces of the tetrahedron radially onto the sphere, we obtain a cover of S^2 by four closed sets, none of which contains a pair of antipodal points.

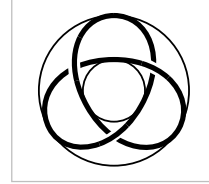
Assuming the higher-dimensional version of the Borsuk-Ulam theorem, the same arguments show that S^n cannot be covered by $n + 1$ closed sets without antipodal pairs of points, though it can be covered by $n + 2$ such sets, as the higher-dimensional analog of a tetrahedron shows. Even the case $n = 1$ is somewhat interesting: If the circle is covered by two closed sets, one of them must contain a pair of antipodal points. This is of course false for nonclosed sets since the circle is the union of two disjoint half-open semicircles.

The relation between the fundamental group of a product space and the fundamental groups of its factors is as simple as one could wish:

Proposition 1.12. $\pi_1(X \times Y)$ is isomorphic to $\pi_1(X) \times \pi_1(Y)$ if X and Y are path-connected.

Proof: A basic property of the product topology is that a map $f: Z \rightarrow X \times Y$ is continuous iff the maps $g: Z \rightarrow X$ and $h: Z \rightarrow Y$ defined by $f(z) = (g(z), h(z))$ are both continuous. Hence a loop f in $X \times Y$ based at (x_0, y_0) is equivalent to a pair of loops g in X and h in Y based at x_0 and y_0 respectively. Similarly, a homotopy f_t of a loop in $X \times Y$ is equivalent to a pair of homotopies g_t and h_t of the corresponding loops in X and Y . Thus we obtain a bijection $\pi_1(X \times Y, (x_0, y_0)) \approx \pi_1(X, x_0) \times \pi_1(Y, y_0)$, $[f] \mapsto ([g], [h])$. This is obviously a group homomorphism, and hence an isomorphism. $\square$

Example 1.13: The Torus. By the proposition we have an isomorphism $\pi_1(S^1 \times S^1) \approx \mathbb{Z} \times \mathbb{Z}$. Under this isomorphism a pair $(p, q) \in \mathbb{Z} \times \mathbb{Z}$ corresponds to a loop that winds p times around one S^1 factor of the torus and q times around the other S^1 factor, for example the loop $\omega_{pq}(s) = (\omega_p(s), \omega_q(s))$. Interestingly, this loop can be knotted, as the figure shows for the case $p = 3, q = 2$. The knots that arise in this fashion, the so-called *torus knots*, are studied in Example 1.24.



More generally, the n -dimensional torus, which is the product of n circles, has fundamental group isomorphic to the product of n copies of $\mathbb{Z}$. This follows by induction on n .

Induced Homomorphisms

Suppose $\varphi: X \rightarrow Y$ is a map taking the basepoint $x_0 \in X$ to the basepoint $y_0 \in Y$. For brevity we write $\varphi: (X, x_0) \rightarrow (Y, y_0)$ in this situation. Then φ induces a homomorphism $\varphi_*: \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$, defined by composing loops $f: I \rightarrow X$ based at x_0 with φ , that is, $\varphi_*[f] = [\varphi f]$. This induced map φ_* is well-defined since a homotopy f_t of loops based at x_0 yields a composed homotopy φf_t of loops based at y_0 , so $\varphi_*[f_0] = [\varphi f_0] = [\varphi f_1] = \varphi_*[f_1]$. Furthermore, φ_* is a homomorphism since $\varphi(f \cdot g) = (\varphi f) \cdot (\varphi g)$, both functions having the value $\varphi f(2s)$ for $0 \leq s \leq 1/2$ and the value $\varphi g(2s - 1)$ for $1/2 \leq s \leq 1$.

Two basic properties of induced homomorphisms are:

- $(\varphi\psi)_* = \varphi_*\psi_*$ for a composition $(X, x_0) \xrightarrow{\psi} (Y, y_0) \xrightarrow{\varphi} (Z, z_0)$.
- $\mathbb{1}_* = \mathbb{1}$, which is a concise way of saying that the identity map $\mathbb{1}: X \rightarrow X$ induces the identity map $\mathbb{1}: \pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$.

The first of these follows from the fact that composition of maps is associative, so $(\varphi\psi)f = \varphi(\psi f)$, and the second is obvious. These two properties of induced homomorphisms are what makes the fundamental group a functor. The formal definition